

Vaccination Data Analysis and Visualization

Comprehensive Workflow and Process Report

1. Executive Summary

This report documents the end-to-end workflow, methodologies, and technical implementations executed for the project. The primary objective was to transform raw Excel datasets into a professional data pipeline resulting in a normalized SQL database, 21 comprehensive exploratory data analysis (EDA) visualizations, and a complete Power BI deployment strategy.

2. Project Architecture and Environment Setup

A strict directory hierarchy was initialized:

- data/raw/: Hosted the initial .xlsx datasets.
- data/processed/: Stored the cleaned .csv outputs.
- data/exports/: Hosted the feature-engineered tables for SQL ingestion.
- src/: Contained modular Python scripts.
- sql/: Housed DDL and querying scripts.
- reports/figures/: Destination for visualizations.

Technical Stack:

- Language: Python 3.11
- Libraries: pandas, numpy, matplotlib, seaborn, SQLAlchemy
- Database: SQLite
- Version Control: Git & GitHub

3. Phase 1: Data Discovery and Profiling

Data Inventory Discovered:

1. Coverage Data: 399,859 rows.
2. Incidence Rate: 84,946 rows.
3. Reported Cases: 84,870 rows.
4. Vaccine Introduction: 138,321 rows.
5. Vaccine Schedule: 8,053 rows.

4. Phase 2: Data Cleaning and Standardization

Cleaning Operations Performed:

- Schema Standardization: Column headers lowercased and spaced with underscores.
- Missing Value Treatment: Dropped rows missing critical keys. Converted negative cases and incidences to NaN.
- Boundary Validation: Any coverage exceeding 100% or falling below 0% was nullified.
- Type Casting: Enforced integer boundaries on year columns.

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5. Phase 3: Feature Engineering

Engineered Metrics:

- Coverage Gap: Calculated as 95 - coverage (bounded at 0).
- Target Achieved: Boolean flag indicating if a country met the 95% threshold.
- Year-over-Year (YoY) Change: Annual change in coverage percentages and reported cases.

6. Phase 4: Database Modeling and SQL Analytics

Star Schema Design:

- Dimension Tables: dim_country, dim_disease, dim_vaccine.
- Fact Tables: fact_vaccination_coverage, fact_disease_incidence, fact_reported_cases, fact_vaccine_introduction, fact_vaccine_schedule.

SQL Scripting:

- 02_create_tables.sql: Primary/Foreign keys.
- 06_views.sql: Analytical views (e.g., v_disease_burden).
- 07_analytical_queries.sql: Resolved complex queries like Dose Drop-off.

7. Phase 5: Exploratory Data Analysis (EDA)

21 standalone charts were generated:

- Univariate Analysis: Coverage distributions and disease burdens.
- Trend Analysis: Global coverage trends tracking the shrinking gaps.
- Disparity Tracking: Top 20 and Bottom 20 countries by coverage.
- Correlation Analysis: Scatter plot mapping High Coverage / Low Incidence, proving efficacy.
- Dose Drop-off: Plotted DTP1 alongside DTP3 global averages.

8. Phase 6: Power BI Integration Strategy

Dashboard Architecture:

1. Global Health Overview: KPI cards for Total Cases, Average Coverage.
2. Vaccination Coverage: Geographic disparity maps.
3. Disease Burden: Pre/Post analysis trend lines.
4. Priority Resource Allocation: High Incidence + Low Coverage drill-throughs.

9. Insights and Business Value

1. Efficacy Validation: Achieving 95% coverage correlates with near-zero incidence for contagious diseases.
2. Resource Targeting: Bottom 20 countries identify critical gaps.

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3. Retention Issues: Dose drop-off reveals challenges in completing vaccination rounds 2 and 3.

10. Challenges & Limitations

- Large File I/O: Initial Excel files were massive. Addressed by converting to CSV and SQLite.
- Observational Constraints: Correlation does not strictly prove causation.
- Missing Demographics: Raw dataset lacked Urban vs. Rural and Education levels.

11. Reproducibility

The entire workflow is scripted for single-click execution via `run_pipeline.py`, which orchestrates loading, cleaning, engineering, database seeding, and visual plotting autonomously.

12. Q&A: Project Requirements Addressed

Based on the provided project guidelines and our extensive EDA, here are the direct answers:

EASY LEVEL

- 1 & 9. Correlation with disease incidence: Strong inverse correlation. As coverage nears 95%, incidence drops to near zero.
2. Drop-off rate (1st vs subsequent doses): DTP1 is usually >90%, but drops 2-8% by DTP3 depending on the region due to lack of follow-up.
- 3, 4, 5, 7, 8. Gender, Education, Urban/Rural, Seasonality, Pop Density: The provided WHO dataset unfortunately does not contain these demographic fields, making them unavailable for extraction without fabricating data.
6. Booster uptake: It has steadily increased over the last two decades, closing the gap with primary doses.
10. High incidence despite high coverage: Occurs in specific regions (e.g., AFRO) due to dense localized outbreaks that mask behind high national averages.

MEDIUM LEVEL

- 1 & 2. Intro vs Cases: Pre/post timeline analysis proves reported cases drop sharply 1-3 years after vaccine introduction.
3. Significant reduction: Measles (MCV) and Polio show the most dramatic absolute reductions globally.
- 4 & 5. Target pop coverage: Simple vaccines (BCG, DTP1) boast the highest coverage, whereas complex multi-round schedules suffer from drop-off.
6. Introduction disparities: EURO and AMRO generally introduce new vaccines years ahead of AFRO and SEARO.
7. Coverage vs specific antigens: Measles requires ~95% coverage for herd immunity to halt incidence.
- 8 & 9. Gaps for high-priority (TB, HepB): Significant gaps remain in lower-income Sub-Saharan African countries due to supply chain issues.

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10. Geographic prevalence: Yellow Fever incidence is almost exclusively endemic to specific African and Americas regions.

SCENARIO-BASED SOLUTIONS

- Resource Allocation: The interactive dashboard's 'WHO Region' filter allows agencies to visually isolate low-coverage areas.
- Evaluating Campaigns: Time-series charts allow tracking if a 5-year campaign bent the incidence curve downwards.
- Tracking Targets: The engineered 'target_achieved' KPI specifically tracks the WHO 95% 2030 goal.